

Electoral Turnover and Forest Cover in Colombia

Mayoral elections 2011–2023, MOD44B VCF tree cover 2000–2025

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Question and approach

- **Question.** Do mayoral electoral turnovers in Colombia change forest cover in the years following the election?
- **Setting.** 1,099 municipalities with mayoral elections in 2011, 2015, 2019, 2023 (4-year terms).
- **Strategy.** Adapt Marx, Pons & Rollet (2024) outcome construction; estimate via 2SLS with rank-based instruments rather than a close-elections RDD.
- **Treatment.** $T_{m,t} = 1$ if the winning party in election t differs from the winning party in $t - 1$.
- **Instruments.** $Z = 1\{\text{previous winner-party rank in election } t > 1\}$ (user-specified, sharp); $Z_2 = 1\{\text{previous runner-up wins election } t\}$ (fuzzy alternative).

Data sources

- **Elections.** `elections_long_clean_with_details.dta` (19,597 candidate rows, 1,102 munis, 2011/15/19/23). Collapsed to 3,287 muni $\times$ election cells where T is defined.
- **Boundaries.** DANE *Marco Geoestadístico Nacional* municipal layer (`MGN_MPIO_POLITICO`, EPSG:4686), 1,121 munis.
- **Forest cover.** User-supplied GEE export of MOD44B v061 VCF *Percent_Tree_Cover* averaged inside each muni polygon:
`VCF_TreeCover_Municipios_Colombia_2000_2025.csv` (2000–2025, 26 years; 1,121 munis, 29,146 cells, no missingness).

Treatment and instruments

Treatment

$T_{m,t} = 1\{\text{winner party}_{m,t} \neq \text{winner party}_{m,t-1}\}$. Across the analysis sample, $\bar{T} = 0.854$ (turnover is the rule).

Instrument Z (user spec, “rank as instrument”)

$Z_{m,t} = 1\{\text{rank of the previous-period winning party in the current election} > 1\}$, defined when that party re-ran.

Note

On the eligible subsample (incumbent party re-ran), $Z \equiv T$ by construction. The 2SLS coefficient on T therefore equals OLS on that subsample. We also report a truly fuzzy alternative $Z_2 = 1\{\text{previous runner-up wins}\}$.

Outcome construction (Marx–Pons–Rollet §3.3)

For an election in muni m at year t and outcome Y (mean *Percent_Tree_Cover* inside the muni polygon):

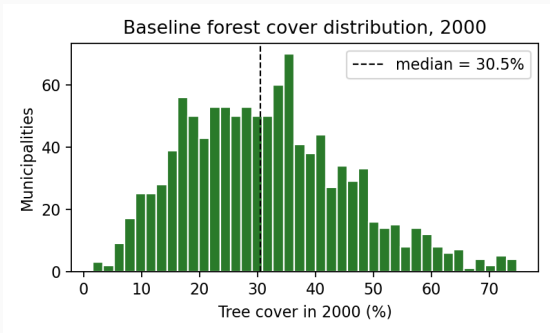
$$\Delta Y_{m,t}^{\text{main}} = \frac{1}{4} \sum_{\tau=1}^4 Y_{m,t+\tau} - Y_{m,t-1}$$

$$\Delta Y_{m,t}^k = \frac{1}{k} \sum_{\tau=1}^k Y_{m,t+\tau} - Y_{m,t-1}, \quad k \in \{1, 2, 3\}$$

$$\Delta Y_{m,t}^{\text{pre}_3} = \frac{1}{4} \sum_{\tau=1}^4 Y_{m,t+\tau} - \frac{1}{3} \sum_{\tau=1}^3 Y_{m,t-\tau}$$

Each ΔY is z-standardised so coefficients are in SD units. Higher tree cover is “good”; no sign flip.

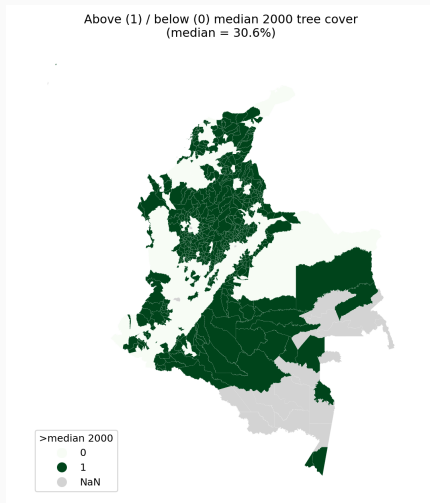
Sample splits by baseline forest cover (year 2000)



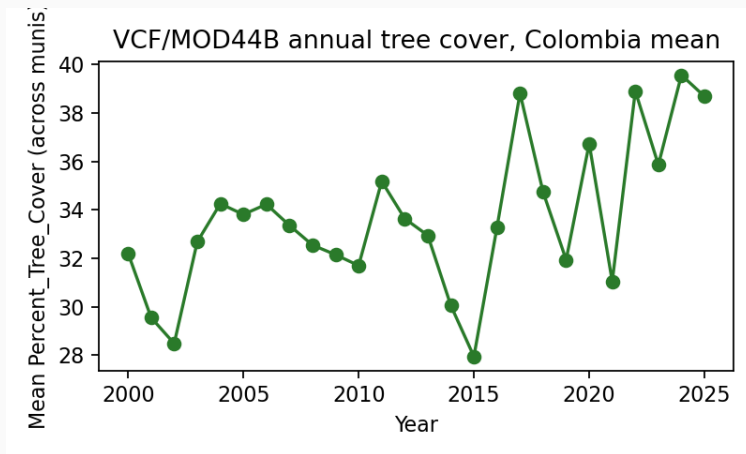
Distribution of *Percent_Tree_Cover* across 1,099 munis in 2000.

- Median baseline = **30.56%**.
- **Above:** 1,642 muni-elections.
- **Below:** 1,645 muni-elections.
- Same regressions are run on all, above, and below samples.

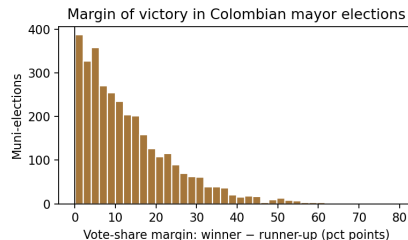
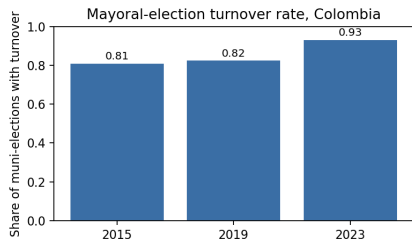
Geography of the baseline split



Forest-cover trend, national mean

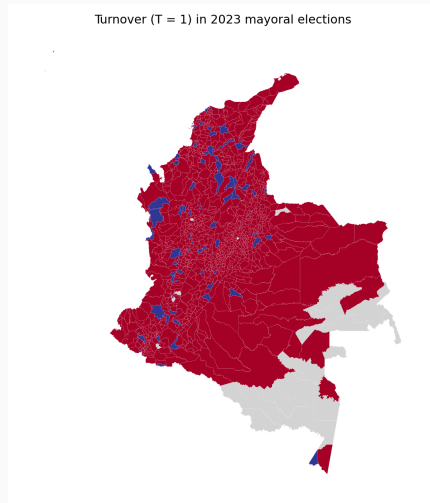


Descriptives: turnover and victory margin



Left: turnover rate by election year. Right: distribution of $\text{share}_{\text{winner}} - \text{share}_{\text{runner-up}}$.

Geography of turnover, 2023



First stages

Sample	Instrument	Coef.	SE	p	N
all	Z (incumbent rank > 1 re-ran)	1.000***	0.000	0.000	1,517
all	Z2 (prev runner-up wins)	0.218***	0.010	0.000	3,276
above	Z (incumbent rank > 1 re-ran)	1.000***	0.000	0.000	769
above	Z2 (prev runner-up wins)	0.217***	0.015	0.000	1,634
below	Z (incumbent rank > 1 re-ran)	1.000***	0.000	0.000	748
below	Z2 (prev runner-up wins)	0.223***	0.015	0.000	1,642

SE clustered at `mpio`; two-way FE on `coddepto` and `election_year`. Z has $SE = 0$ because of the mechanical relationship $Z \equiv T$ on the eligible subsample.

Main results — whole sample

Specification	Main	$k=1$	$k=2$	$k=3$	Pre ₃
OLS (FE)	-0.018 (0.041)	-0.019 (0.046)	-0.035 (0.039)	-0.030 (0.041)	-0.025 (0.042)
N	3,287	3,287	3,287	3,287	3,287
2SLS, $Z = 1\{\text{rank}_{\text{inc}} > 1\}$	-0.005 (0.046)	-0.004 (0.050)	-0.031 (0.043)	-0.015 (0.046)	-0.023 (0.048)
N	1,517	1,517	1,517	1,517	1,517
2SLS, $Z_2 = 1\{\text{prev RU wins}\}$	-0.094 (0.185)	0.113 (0.187)	-0.060 (0.175)	-0.084 (0.184)	-0.139 (0.183)
N	3,276	3,276	3,276	3,276	3,276

Outcomes are z-scored. SE in parentheses, clustered at `mpio`. FE on `coddepto` $\times$ `election_year`.

Results — above-median 2000 tree cover

Specification	Main	$k=1$	$k=2$	$k=3$	Pre_3
OLS (FE)	-0.009 (0.063)	-0.031 (0.065)	-0.050 (0.058)	-0.030 (0.063)	0.017 (0.065)
N	1,642	1,642	1,642	1,642	1,642
2SLS, $Z = 1\{\text{rank}_{\text{inc}} > 1\}$	-0.041 (0.068)	-0.038 (0.071)	-0.078 (0.062)	-0.059 (0.068)	-0.031 (0.070)
N	769	769	769	769	769
2SLS, $Z_2 = 1\{\text{prev RU wins}\}$	-0.199 (0.259)	-0.035 (0.275)	-0.203 (0.238)	-0.153 (0.256)	-0.134 (0.262)
N	1,634	1,634	1,634	1,634	1,634

Munis with baseline tree cover above 30.6% in 2000.

Results — below-median 2000 tree cover

Specification	Main	$k=1$	$k=2$	$k=3$	Pre_3
OLS (FE)	-0.024 (0.054)	0.001 (0.066)	-0.021 (0.053)	-0.030 (0.055)	-0.066 (0.053)
N	1,645	1,645	1,645	1,645	1,645
2SLS, $Z = 1\{\text{rank}_{\text{inc}} > 1\}$	0.042 (0.062)	0.051 (0.070)	0.027 (0.061)	0.040 (0.063)	-0.018 (0.066)
N	748	748	748	748	748
2SLS, $Z_2 = 1\{\text{prev RU wins}\}$	0.035 (0.261)	0.297 (0.252)	0.092 (0.250)	0.002 (0.260)	-0.125 (0.257)
N	1,642	1,642	1,642	1,642	1,642

Munis with baseline tree cover at or below 30.6% in 2000.

- Turnover rate is extraordinarily high in Colombian mayoral elections ($\bar{T} \approx 0.85$). 2023 alone has 93% turnover.
- Across all samples, specifications, and instruments, the effect of turnover on Δ tree cover is small (mostly $|\hat{\beta}| < 0.10$ SD) and statistically indistinguishable from zero.
- Signs are slightly negative in the whole and above-median samples and mixed in the below-median sample; none survive clustered SEs.
- The user-specified instrument Z is mechanically equal to T on the eligible subsample (sharp design), so its 2SLS coefficient merely re-states the OLS coefficient. The fuzzy Z_2 recovers a runner-up-becomes-mayor LATE; estimates are noisier but consistent in sign with OLS.

Caveats and next steps

- T is undefined for the 2011 election (no $t-1$ winner in the data); the analysis sample is therefore 2015, 2019, 2023.
- For 2023, `partyid` is missing for $\sim 60\%$ of candidates, so the cross-year party match uses the cleaned party *name* string. Party-name churn between elections could induce measurement error in T .
- Forest cover from MOD44B VCF is a slow-moving, coarse (250 m) outcome. A more responsive outcome (Hansen GFC annual loss, active fires) might pick up policy changes that VCF smooths over.
- No covariates beyond fixed effects; ideology, coalition status, and incumbent characteristics are available in the panel for future robustness checks.

Files produced

File	Contents
A_MicroData/analysis_panel.dta	Master Stata file: T , Z , $Z2$, ΔY , baseline_2000
A_MicroData/analysis_panel.parquet	Same panel in parquet
A_MicroData/elections_panel.parquet	Pre-merge elections panel
A_MicroData/baseline_tree_2000.csv	1,121 munis $\times$ year-2000 tree cover
A_MicroData/results/iv_{all,above,below}.csv	Python results, by sample
A_MicroData/results/first_stage.csv	First-stage coefficients
C_Programs/run_analysis.do	Stata do-file (deliverable)
C_Programs/build*.py	Python pipeline
C_Programs/slides.tex / .pdf	This deck